

RÉSUMÉ

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EDUCATION AND QUALIFICATIONS

School of Engineering, Georgia Institute of Technology, Atlanta, GA

Aug 2023 – Jun 2028

Doctor of Philosophy (Ph.D.), Electrical and Computer Engineering. CGPA: 3.83/4.0

- Graduate Research Assistant – AI Institute for Advances in Optimization (AI4OPT) | Energy Division
- **Georgia Tech Supervisors:** Prof. Pascal Van Hentenryck and Prof. Daniel Molzahn
- **External Industry and Academic Mentors:** Dr. Russell Bent (LANL) and Prof. Parikshit Pareek (IIT Roorkee)

School of Engineering, Georgia Institute of Technology, Atlanta, GA

Aug 2023 – May 2025

Master of Science (MS.), Electrical and Computer Engineering. CGPA: 3.80/4.0

- Minor: Industrial and Systems Engineering (ISyE) – *Regression Analysis, Convex Optimization & Machine Learning*
- Coursework: **Power Systems** (*Protection, Cybersecurity, Operation & Control*), **Statistical Methods**, **Linear Systems & Control**

Engineering Department, Ashesi University, Accra, Ghana

Sep 2018 – Jun 2022

Bachelor of Science (B.Sc.), Electrical and Electronics Engineering. Cum Laude

- B.Sc. Project: Overall Best Thesis – Engineering Department [Published in IEEE Xplore | Citations: 14+]

RESEARCH EXPERIENCE

Graduate Research Consultant, Austin, TX

Oct 2025 – Present

Electric Reliability Council of Texas (ERCOT), *Risk Assessment of Texas Grid Using AC SCOPF*

- Collaborating with ERCOT engineers on machine learning approaches for Security-Constrained AC Optimal Power Flow (AC SCOPF) and greater than N-3 contingency analysis, using real-world operational data from the Texas grid
- Developing a self-supervised primal-dual learning neural-network that incorporates AC non-linearities into preventive SCOPF pipelines, with the goal of enhancing grid reliability under large-scale system conditions (1000+ buses)

Graduate Research Intern, Los Alamos, NM

May 2025 – Present

Los Alamos National Laboratory (LANL), *End-to-End Learning for AC Feasibility Restoration in DCOPF*

- Designing an LMP-preserving DCOPF-ACPF pipeline that learns DCOPF parameters and generator participation factors, and computes pseudo-LMPs from AC-PF Jacobian sensitivities to keep AC prices aligned with DC outcomes
- Implementing implicit-function gradients—KKT adjoints for DC-OPF and a single transpose-Jacobian solve for AC-PF—to enable end-to-end training in Python with NumPy/SciPy, PyTorch, CVXPY, and OSQP/GUROBI
- Prototyping fast feasibility restoration with smooth voltage-reactive power handling, while exploring tap/shunt policies to cut recalculations and runtime on large test cases by 90%, compared to conventional approaches

Graduate Research Intern, Los Alamos, NM

Sept 2024 – Aug 2025

Los Alamos National Laboratory (LANL), *Exploring DCOPF and ACPF Formulations for AC Feasibility*

- Engineered a unified DCOPF→ACPF recovery pipeline combining four loss-aware DC models with AC feasibility restoration techniques including distributed slack allocation, PV/PQ bus switching, and voltage-informed initialization
- Evaluated the pipeline on 10,000+ stochastic dispatch scenarios across large-scale power systems, demonstrating feasibility restoration in the major cases and reducing inequality constraint violations by 3–5× under extreme loading
- Reduced Newton-Raphson convergence iterations by ~50% in stressed test cases and eliminated active/reactive power violations across challenging industry-grade networks, including PEGASE and RTE benchmark systems
- Achieved 75% lower mean absolute error and 93% cost difference reduction compared to traditional single-slack methods, showing the effectiveness of incorporating line losses and distributed control in DCOPF formulations

Graduate Research Assistant, Atlanta, GA

Apr 2025 – Oct 2025

AI Institute for Advances in Optimization (AI4OPT), *Volt-VAR Optimization for Large-Scale Power Systems*

- Designed a transmission-scale Volt/VAR Optimization framework that co-optimizes OLTC tap ratios and capacitor-bank modules under AC power-flow physics, implementing a relax-round-resolve pipeline, that encodes device limits
- Formulated operations-aligned objectives that hold voltage and VAR setpoints while penalizing active-power redispatch and preserving economic dispatch, integrating ACOPF/market references for AC-feasible setpoints
- Performed an extensive review of transmission-scale Volt/VAR and discrete ACOPF (MINLP, relax-round-resolve, cutting-plane/OA, homotopy, metaheuristics, interior-point); pinpointing gaps in (i) scalable AC-feasible tap/shunt schedules, (ii) realistic device step modeling, and (iii) coordination with market dispatch and loss/price sensitivities

Graduate Research Assistant, Atlanta, GA

Dec 2024 – Present

Georgia Institute of Technology (Georgia Tech), *Dynamic State Estimation in Current Transformer Merging Units*

- Trained and evaluated four regression models—Multiple Linear, Ridge, Lasso, and Elastic Net—on over 52,000 observations, using an 80/20 train-test split and environmental predictors like temperature, humidity, and wind speed
- Applied statistical preprocessing techniques (normalization, standardization, and correlation filtering), revealing strong intra-zone correlations (e.g., Zone 1 to overall demand: $\rho = 0.959$) and weak relevance of wind speed and diffuse flows
- Achieved best generalization with Lasso Regression, reducing Mean Absolute Error to 134.79 and avoiding overfitting observed in MLR (MAE = 3.04) by balancing model complexity with predictive accuracy, stability, and robustness

Graduate Research and Teaching Assistant, Atlanta, GA

Oct 2024 – Present

Georgia Institute of Technology (Georgia Tech), *Design Fundamentals (ECE 3011)*, *Digital Design Lab (ECE 2031)*

- Introduced faculty and students to Electronic Design Automation (EDA) as an alternative to Eagle and Altium, and created a full set of lectures and video guides to support final-project Printed Circuit Board (PCB) design
- Leading the development of MPower Boards (MPBs), integrating 3-D printing, LED technology, and real-time simulations to create a physical, interactive model of power grids for enhanced educational and research
- Advancing commercialization with Georgia Tech's Create-X program by building a portable, battery-powered device, scoping IP for a provisional patent, and piloting IEEE-14/30/39 classroom demos, to visualize grid line faults

SKILLS

- **Languages:** English (Native), Sign Language, C++, MATLAB, PHP, MySQL, HTML, CSS, R, Python, and Julia
- **Tools:** PowerModels, MATPOWER, PG-LIB, Pandapower, PowerWorld, PyPSA, SciPy, CVXPY, PyTorch, Git

HONORS AND AWARDS

Argonne National Lab's Training Program on Extreme-Scale Computing (1 of 70 Nationwide), IL, USA	<i>Jul 2026</i>
Create-X Entrepreneurship Program with \$5000 Seed Funding for Power Systems Education, GA USA	<i>Feb 2026</i>
GT's Up with the White and Gold 'Rising Star Award': Center for Student Engagement, GA USA	<i>Apr 2026</i>
Center for Nonlinear Studies Grid Science Winter School & Conference: Travel Grant Winner, NM, USA	<i>Jan 2025</i>
Argonne Department of Energy's CyberForce® Competition: Travel Grant Winner, IL, USA	<i>Nov 2024</i>
International Solid-State Circuits Conference: Travel Grant Winner (1 of 4, 1 st Year ECE PhDs), CA USA	<i>Jan 2024</i>
National Society of Black Engineers Conference: NSBE50 Scholarship, GA USA	<i>Mar 2024</i>
Silver Medalist: Poster Presentation, International Genetically Engineered Machine (<i>Competition</i>), France	<i>Oct 2022</i>

RECENT JOURNAL/ CONFERENCE PUBLICATIONS

- **M.A. Boateng**, R. Bent, S. Misra, P. Pareek, P. Van Hentenryck, and D. Molzahn, "Towards AC Feasibility of DCOPF Dispatch," *Electric Power Systems Research*, 2026. To appear. Presented at the 24th Power Systems Computation Conference (PSCC), June 8–12, 2026 | [Acceptance rate: **34.6%** in 2026].
- S. Tong, **M.A. Boateng**, M. Tanneau, and P. Van Hentenryck, "Volt/VAR Optimization in Transmission Networks with Discrete-Control Devices," *Electric Power Systems Research*, 2026. To appear. Presented at the 24th Power Systems Computation Conference (PSCC), June 8–12, 2026 | [Acceptance rate: **34.6%** in 2026].
- R. Asiamah, S. Talkington, **M. Boateng**, M. Vanin, F. Geth, and D. Molzahn, "Classifying Reactive Power Control Laws of Behind-the-Meter Solar Photovoltaic Inverters," *Proc. IEEE Kansas Power and Energy Conf. (KPEC)*, 2025.

PRESENTATIONS/ TALKS

AI Institute for Advances in Optimization, Industry Partners Meeting, GA, USA	<i>Nov 2025</i>
Los Alamos National Lab, Lightning Talks T-5 Division, NM, USA	<i>Jul 2025</i>

REVIEWED CONFERENCE/ JOURNAL PAPERS

- 2025 24th *Power Systems Computation Conference (PSCC)*, Track: AC-Power Flow, and Volt-Var Optimization
- 2025 *Electric Power Systems Research Journal* (Elsevier, Impact Factor: 4.2; CiteScore: 8.2). Track: Neural Networks
- 2025 IEEE *Kansas Power and Energy Conference (KPEC)*, Track: Solar Power, Transmission and Distribution Systems

LEADERSHIP, ACTIVITIES, AND COMMUNITY INVOLVEMENT

Chief of Staff: Georgia Tech Graduate Student Government Association , GA, USA	<i>May 2026 – Present</i>
President: Fellowship of Christian Graduate Students , GA, USA	<i>Jul 2025 – May 2026</i>
Vice President: Georgia Tech Ghanaian Students Association , GA, USA	<i>Jun 2025 – May 2026</i>
Chair, Special Projects and Programs: African Graduate Students Committee , GA, USA	<i>May 2025 – May 2026</i>
Co-Captain: International Genetically Engineered Machine Competition (iGEM) , Paris, France	<i>Jun – Oct 2022</i>
Editor-in-Chief: Ashesi Science Engineering Entrepreneurship Design (SEED) Journal , Ghana	<i>2022 – 2023</i>